Sayand P

ML Engineer — MLOps · LLM Systems · Production Python

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I ship machine learning that runs unattended: scheduled retraining, model registries, drift monitoring and test suites that fail loudly. Seeking an ML Engineer role, remote.

PROJECTS

Kerala Crop Price Forecaster

Live Telegram bot (Malayalam) · github.com/sayandp/kerala-crop-forecaster

- Built and deployed a free Malayalam price service for farmers: today's mandi price for 17 crops across every Kerala market, a 7-day outlook, and user-set price alerts. Refreshes itself daily; runs entirely on free tiers at ₹0/month.
- Trained LightGBM on 750,000 price records since 2018 and found it could not beat a naive "tomorrow = today" rule at 1–7 day horizons; shipped the naive forecast with a calibrated interval (~80% coverage) and gated the model to the horizons where it demonstrably wins.
- Automated the lifecycle end to end: scheduled ingest and retraining through GitHub Actions, experiment tracking and model registry in MLflow, drift monitoring on incoming data.

Stack: Python, LightGBM, MLflow, GitHub Actions, Postgres, FastAPI, Next.js, Telegram Bot API

Agentic Data Intelligence Platform

PG Diploma capstone · github.com/sayandp/agentic-data-intelligence-platform

- Designed a nine-agent pipeline that ingests CSV, Excel, SQL and REST behind one data contract, detects schema drift and distribution shifts, repairs safe faults automatically and generates grounded analytics reports. 1,075 backend tests, 72 end-to-end browser tests.
- Separated reasoning from authority in three layers — deterministic detection, LLM diagnosis under a closed enum schema, rule-gated action — giving 11/12 correct diagnoses and zero incorrectly applied high-risk repairs across 10 swept thresholds.
- Measured that 82% of the model's self-reported confidence landed at exactly 0.95 within a 0.08-wide band, proving the threshold did no routing work; also cut a 1.07M-row run from 16 min to 2 min 36 s and shipped PII redaction with a full egress audit.

Stack: Python, LangGraph, FastAPI, React, Postgres, Playwright, Gemini

Kerala Flood Risk Intelligence System

github.com/sayandp/kerala-flood-risk-intelligence-system

- Property-level flood risk from terrain rather than weather: Random Forest on 992 spatially stratified samples using elevation, slope, river distance, catchment area and historical paddy land from Copernicus DEM, Sentinel-1, Dynamic World and 159,168 OpenStreetMap waterway features.
- 91.96% accuracy, 0.975 ROC-AUC and 94.83% ± 1.92% on 5-fold cross-validation, with SHAP explanations per prediction; 12 physics-based hydrology rules constrain the model on geographic edge cases, and 10/10 validation locations classified correctly.

Stack: Python, scikit-learn, SHAP, rasterio, FastAPI, React, Leaflet

TECHNICAL SKILLS

Languages	Python, SQL, R
MLOps	MLflow, GitHub Actions, Docker, scheduled retraining, model registry, drift monitoring, Evidently
Serving	FastAPI, Postgres, Next.js, Streamlit, Render, Vercel, Telegram Bot API
Machine learning	scikit-learn, XGBoost, LightGBM, time-series forecasting, SHAP, cross-validation, hyperparameter tuning
LLM systems	LangGraph, tool-calling agents, RAG, structured output under closed schemas, evaluation harnesses
Testing	pytest, Playwright, load testing, falsification scripts for safety guarantees

EDUCATION

PG Diploma, Data Science & Analytics — University of Calicut 2025 – 2026

B.Sc. Statistics with Data Science — University of Calicut 2021 – 2024 · First Class

ALSO BUILT

Sahayi — bilingual retrieval assistant with citation-or-abstain evaluation · India GDP Nowcaster · E-commerce SQL analytics dashboard · Loan default risk engine · Four ML apps deployed on Streamlit Cloud · Census OCR pipeline (EasyOCR, FastAPI, React)